



Unit 3 · Outcome 1 · Maintaining Biodiversity

Simpson's Index of Diversity

KK07

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. 07 Simpson's Index of Diversity Counting species tells you how rich a place is

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Simpson's Index of Diversity

Unit 3 · Outcome 1 · Key Knowledge 07 Simpson's Index of Diversity Counting species tells you how rich a place is. But two forests with the same species list can be wildly different. Simpson's Index of Diversity captures what a headcount misses — evenness — by asking one elegant question: pick two individuals at random; how likely is it they're different species? Live · probability engine building the sample plots... Two plots, same species, same totals — watch the draws separate them. ► Run random draws 5 tally Each draw grabs two individuals. Different s

01 Three ways to measure species diversity

01 Three ways to measure species diversity The study design names three tools. They answer different questions, and the exam expects you to know what each one can and can't see. Species richness The total number of different species in a defined area. Count the species, ignore how many of each. Quick and intuitive. a count: e.g. "12 species" Endemism A species is endemic if it's found naturally nowhere else on Earth. High endemism means a place holds irreplaceable diversity — losing it is a global, not local, loss. a property: "found only here" Simpson's

02 Why a species count isn't enough

02 Why a species count isn't enough Drag the dominance slider. Both communities below always have the same 5 species and the same 100 individuals — only the spread changes. Richness never moves. Watch what SID does. Interactive · evenness raising the canopy... Five species · 100 individuals · richness fixed at 5 Dominance of top species even Species richness 5 never changes Simpson's Index (SID) 0.80 high diversity Top species share 20% of all individuals ▲ The takeaway Slide all the way to one dominant species and SID collapses toward 0 — even though rich

03 The formula, decoded

◆ COMMAND WORDS

03 The formula, decoded VCAA gives you the formula in the exam — your job is using it without slipping. Hover or tap each part to see what it means. $SID = 1 - \frac{\sum n(n-1)}{N(N-1)}$ n the number of individuals of each species — you work out $n(n-1)$ for every species, then add them up (that's the $\sum$) N the total number of individuals of all species combined — not the number of species! $1 -$ flips the score so that higher = more diverse, matching intuition SID a value from 0 to 1 = the probability two random individuals are different species ✗ The fraction al

04 Worked example — two plots at Kestrel Ridge

04 Worked example — two plots at Kestrel Ridge A Year 12 class surveys ground-layer plants in two 4 m^2 quadrats at the fictional Kestrel Ridge Reserve (box-ironbark woodland, central Victoria). Both quadrats hold the same 5 species and the same 56 individuals. Use the toggle to switch sites and watch the calculation rebuild. Site A · intact gully Site B · grazed edge Site A · intact gully laying out the quadrat... Each marker = one plant, coloured by species. Click a species in the table to highlight it. Species n $n-1$ $n(n-1)$ Totals 56 — 598 Total the in

Species	n	$n-1$	$n(n-1)$
Totals	56	—	598

05Reading the number

05 Reading the number A bare "SID = 0.81" earns little. Marks come from interpreting it in context and comparing. What the value means SID near 0 Low diversity — one or few species dominate (monoculture, degraded site) SID mid (~0.5) Moderate — some spread but uneven SID near 1 High diversity — many species, evenly abundant Why higher diversity matters A higher SID generally signals an ecosystem that is more resilient and more stable : more species and more even abundances mean redundancy in roles, so the system can absorb disturbance (disease, drought,

SID near 0	Low diversity — one or few species dominate (monoculture, degraded site)
SID mid (~0.5)	Moderate — some spread but uneven
SID near 1	High diversity — many species, evenly abundant

06Sort it out — richness, evenness, endemism or SID?

◇ ACTIVE RECALL

06 Sort it out — richness, evenness, endemism or SID? Drag each statement to the measure it describes. (On touch, tap a chip then tap a bin.) Richness Evenness Endemism SID s

07Command-word decoder

◆ COMMAND WORDS

07 Command-word decoder The verb in the question tells you exactly what to deliver. Same topic, very different answers. Calculate Compare Explain Describe Identify / State Justify / Evaluate

08Exam traps

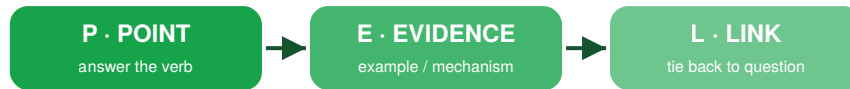
⚠ EXAM TRAPS

08 Exam traps The marks lost on this dotpoint are predictable. Open each to defuse it. ! Forgetting the "1 -" > Students stop at $D = \sum n(n-1)/N(N-1)$ and call it the answer. That's the probability of drawing the same species — the opposite direction. Fix: always finish with $SID = 1 - D$. Sanity check: a clearly diverse site should give a high number. ! Using number of species for N > N is the total number of individuals , not the number of species. Putting N = 5 (the richness) instead of N = 56 wrecks the calculation. Fix: N = the sum of the whole abundance

09P·E·L writing trainer

✎ P·E·L WRITING

09 P·E·L writing trainer For "explain" and "compare" answers, build each point as Point → Explain → Link . Step through a model response below. P Point E Explain L Link ◀ Back Next ▶ Prompt: "Compare the diversity of Site A and Site B and explain the difference." (3 marks)



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

10Good vs weak — the same question, two answers

✗ WEAK ANSWER

Weak response 1 / 3 Site A has more species so it is more diverse. It has a higher number which means it is better. Site B is worse because it has less. So A wins.

✓ STRONG ANSWER & MARKING

Why it loses marks: the sites have equal richness, so "more species" is wrong and self-contradicting. "Better/worse/wins" isn't scientific language. No mention of evenness or what SID m



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

11Practice & self-mark

★ PRACTICE & SELF-MARK

11 Practice & self-mark Five graded questions, 20 marks total. Attempt each on paper first, reveal the model answer, then award yourself marks — the dock tracks your total. ▶ How to self-mark honestly Award a mark only if your wording carries the same idea as a bolded mark-point — not just a near-miss. For calculations, full marks need the working and the final SID with the "1 -" applied.

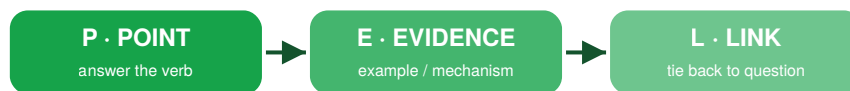
12Key takeaways

♦ RECAP / CHEAT SHEET

12 Key takeaways ♦ Three measures: species richness (how many species), endemism (found only here), and SID (richness + evenness in one 0–1 number). ♦ $SID = 1 - \frac{\sum n(n-1)}{N(N-1)}$, where n = individuals of each species and N = total individuals. The "1 -" makes higher = more diverse. ♦ It's a probability: the chance two random individuals are different species. ♦ Evenness is the point: same richness can give very different SID. Dominance by one species drives SID toward 0. ♦ Interpret, don't just state: value → compare → meaning (diversity) → consequence (r

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Site A has more species so it is more diverse. It has a higher number which means it is better. Site B is worse because it has less. So A wins. Why it loses marks: the sites have equal richness, so "more species" is wrong and self-contradicting. "Better/worse/wins" isn't scientific language. No mention of evenness or what SID measures.

✓ STRONG / MODEL ANSWER

Both sites have the same species richness (5) and the same total individuals, so the difference must come from evenness. In Site A the individuals are spread evenly across the five species, whereas in Site B one species dominates the count. Because Simpson's Index of Diversity rewards an even spread, two randomly chosen individuals in Site A are more likely to be different species, giving Site A the higher SID and indicating greater diversity. Why it scores: identifies equal richness, names evenness as the cause, explains the dominance in B, and links it back to what SID actually measures.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.